# Dynamic Posted Prices for Additive Combinatorial Auctions: A Verified Partial Result for Problem W4408830023

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**Abstract**

Brustle, Correa, Duetting, Ezra, Feldman, and Verdugo asked whether fully adaptive item posted prices for XOS or submodular combinatorial auctions have $(1-\varepsilon)$ competition complexity $\Theta(\log\log(1/\varepsilon))$, matching the single-item block-threshold phenomenon. This note verifies and cleans up a partial result: the desired bound is true for the additive subclass. The proof is a direct but exact reduction of an additive combinatorial auction to a direct sum of single-item block-threshold problems. Since one-item instances are additive, this also gives a matching lower bound for the additive subclass.

The result below is therefore genuine partial progress, but it is not a solution of the open XOS/submodular problem. The proof uses itemwise separability of demand, which fails even for unit-demand valuations. To make this boundary precise, the note also records a general XOS revenue-plus-utility certificate with continuation prices and a small minimax calculation showing that the standard balanced-price coefficient template cannot by itself improve on exponential-in-the-number-of-blocks convergence.

## 1 Status and relation to the open problem

The open problem concerns combinatorial auctions with a finite item set $M$ and $n$ buyer types. In a single block one buyer of each type arrives, with independent valuations $v_i \sim F_i$. In the $k$-block resource-augmentation instance, there are $k$ independent copies of this type sequence. A dynamic posted-price policy may update item prices after each arrival and after each purchase. The benchmark is the expected offline optimum in one block,

$$\mathbb{E}[\mathrm{OPT}_1] = \mathbb{E}\left[\max_{(S_1,\ldots,S_n)\in\mathcal{P}(M)} \sum_{i=1}^{n} v_i(S_i)\right],$$

where $\mathcal{P}(M)$ denotes feasible disjoint allocations of the items. The open question asks whether for XOS or submodular valuations one can obtain

$$\mathbb{E}[\mathrm{SW}_k] \geq (1-\varepsilon)\mathbb{E}[\mathrm{OPT}_1] \qquad \text{with} \qquad k = \Theta(\log\log(1/\varepsilon)).$$

The result proved here establishes this statement only for additive valuations. Additive valuations form a subclass of both XOS and submodular valuations, so the theorem is inside the model of the open problem. However, a theorem for a subclass is not a theorem for the full XOS or submodular class. The correct status is therefore:

$\boxed{\text{solution for additive valuations; partial progress for the XOS/submodular open problem.}}$

Throughout, expectations are assumed finite. Equivalently, all claims may be read after truncating valuations at level $H$ and then sending $H \to \infty$ by monotone convergence.

## 2 Model and preliminaries

There is a finite item set $M$ and buyer types $i \in [n]$. In block $b \in [k]$, buyer $(b, i)$ has valuation $v_i^{(b)} \sim F_i$, independently across $i$ and $b$. Before an arrival, the mechanism posts a nonnegative price $p_j$ for each currently unsold item $j$. The buyer selects a utility-maximizing bundle from the remaining items. The welfare obtained by the posted-price allocation on $k$ blocks is denoted $\mathrm{SW}_k$.

**Definition 2.1** (Additive valuations)**.** *An instance is additive if, for every type $i$, there are nonnegative random variables $(X_{i,j})_{j\in M}$ such that*

$$v_i(S) = \sum_{j\in S} X_{i,j} \qquad (S \subseteq M).$$

*The vector $(X_{i,j})_{j\in M}$ may have arbitrary internal correlation for a fixed type $i$. The vectors for different types are independent.*

For additive valuations the one-block offline optimum decomposes over items:

$$\mathrm{OPT}_1(X) = \sum_{j\in M} \max_{i\in[n]} X_{i,j}. \tag{1}$$

Indeed, after additivity removes all cross-item interactions, each item can be assigned independently to a highest-value type.

**Definition 2.2** (XOS valuations)**.** *A valuation $v : 2^M \to \mathbb{R}_{\geq 0}$ is XOS if there is a family of nonnegative additive clauses $(a^\ell)_{\ell\in\mathcal{L}}$ such that*

$$v(S) = \max_{\ell\in\mathcal{L}} \sum_{j\in S} a_j^\ell \qquad (S \subseteq M).$$

*If $S$ is fixed, a clause $a$ supports $S$ if $v(S) = \sum_{j\in S} a_j$ and $v(T) \geq \sum_{j\in T} a_j$ for every $T \subseteq M$.*

Every additive valuation is XOS. Every nonnegative additive valuation is also submodular, so the additive theorem below applies to a common subclass of the classes in the open problem.

**Tie-breaking.** For the additive reduction it is convenient to avoid zero-surplus item ties. This can be done in either of two standard ways. One may assume atomless item-value distributions, or one may add independent continuous perturbations of size at most $\eta$ to the posted prices and then let $\eta \downarrow 0$. The perturbation makes zero surplus occur with probability zero, and finite expectation lets the welfare guarantee pass to the limit. The XOS certificate in Section 5 does not need a tie-breaking convention, because it uses only the buyer's maximum utility value and the revenue of whatever bundle is chosen.

## 3 The single-item input

The proof uses the single-item result of Brustle, Correa, Duetting, Ezra, Feldman, and Verdugo [1]. We need only the following consequence.

**Fact 3.1** (Single-item block thresholds)**.** *There is a sequence $\beta_k \uparrow 1$ with*

$$1 - \beta_k \leq C \exp(-a^k)$$

*for universal constants $C < \infty$ and $a > 1$, such that the following holds. For every independent sequence of nonnegative random variables $Y_1, \dots, Y_n$, there are block thresholds $\tau_1, \dots, \tau_k$ for the $k$ independent blocks of the sequence such that the threshold rule that accepts the first value in block $b$ exceeding $\tau_b$ has expected reward at least*

$$\beta_k \, \mathbb{E}\left[\max_{i \in [n]} Y_i\right].$$

*Moreover, the worst-case one-item $(1-\varepsilon)$ competition complexity is $\Theta(\log\log(1/\varepsilon))$.*

The lower-bound part may be interpreted against all online stopping rules; it therefore also applies to posted-price policies, which observe no more information than an online stopping rule that sees the realized values.

# 4 The additive theorem

**Theorem 4.1** (Upper bound for additive combinatorial auctions)**.** *For every additive combinatorial auction instance and every $k \geq 1$, there is a dynamic item posted-price policy such that*

$$\mathbb{E}[\mathrm{SW}_k] \geq \beta_k \, \mathbb{E}[\mathrm{OPT}_1],$$

*where $\beta_k$ is the single-item sequence from Fact 3.1. Consequently the additive subclass has $(1-\varepsilon)$ competition complexity $O(\log\log(1/\varepsilon))$ under dynamic posted prices.*

*Proof.* Fix an additive instance. For each item $j \in M$, apply Fact 3.1 to the independent type values

$$X_{1,j}, X_{2,j}, \dots, X_{n,j}.$$

This gives item-specific block thresholds $\tau_{1,j}, \dots, \tau_{k,j}$ such that the single-item threshold process for item $j$ has expected reward at least

$$\beta_k \, \mathbb{E}\left[\max_{i \in [n]} X_{i,j}\right]. \tag{2}$$

The posted-price policy is itemwise: in block $b$, every unsold item $j$ is posted at price

$$p_{b,i,j} = \tau_{b,j}$$

for every type $i$. The price depends on the block and the item, and the availability of items is updated after every purchase.

Consider buyer $(b, i)$ facing the remaining item set $R$. Its utility from a bundle $S \subseteq R$ is

$$v_i^{(b)}(S) - \sum_{j \in S} \tau_{b,j} = \sum_{j \in S} (X_{i,j}^{(b)} - \tau_{b,j}).$$

Thus the buyer's utility maximization problem separates over items. Except on the zero-probability tie event handled above, the buyer purchases precisely those remaining items for which $X_{i,j}^{(b)} > \tau_{b,j}$.

Therefore, for each fixed item $j$, the sale process induced by the multi-item posted-price mechanism is exactly the single-item block-threshold process with thresholds $\tau_{1,j}, \dots, \tau_{k,j}$. Let $A_{k,j}$ be the value contributed to welfare by item $j$, with $A_{k,j} = 0$ if item $j$ is not sold. By (2),

$$\mathbb{E}[A_{k,j}] \geq \beta_k \, \mathbb{E}\left[\max_{i \in [n]} X_{i,j}\right].$$

Finally, welfare is additive over sold items, so linearity of expectation and (1) give

$$\begin{aligned}\mathbb{E}[\mathrm{SW}_k] &= \sum_{j\in M} \mathbb{E}[A_{k,j}] \\ &\geq \beta_k \sum_{j\in M} \mathbb{E}\left[\max_{i\in[n]} X_{i,j}\right] \\ &= \beta_k\, \mathbb{E}[\mathrm{OPT}_1].\end{aligned}$$

Since $1-\beta_k \leq C\exp(-a^k)$, choosing $k = O(\log\log(1/\varepsilon))$ makes $\beta_k \geq 1-\varepsilon$. $\square$

**Theorem 4.2** (Lower bound for the additive subclass)**.** *The worst-case* $(1-\varepsilon)$ *competition complexity of additive combinatorial auctions under dynamic posted prices is*

$$\Omega(\log\log(1/\varepsilon)).$$

*Together with Theorem 4.1, this gives*

$$\mathrm{CC}^{\mathrm{dyn}}_{\mathrm{additive}}(1-\varepsilon) = \Theta(\log\log(1/\varepsilon)).$$

*Proof.* Restrict to instances with $|M| = 1$. Then additive, XOS, submodular, and single-item valuations coincide. A dynamic posted-price policy for one item is a stopping rule, possibly less informed than a prophet-inequality stopping rule that directly observes the realized values. Hence the single-item lower bound in Fact 3.1 embeds unchanged into the additive class. The upper bound is Theorem 4.1. $\square$

## 5 A verified XOS continuation-price certificate

The additive proof succeeds because the buyer's demand separates item by item. For general XOS valuations this separation is false. Nevertheless, one can still write a useful revenue-plus-utility certificate. The certificate below is not strong enough, as currently analyzed, to solve the open problem; it is included to isolate exactly what remains missing.

Fix an XOS instance. For every valuation profile $\mathbf{v} = (v_1, \ldots, v_n)$, choose an offline optimal allocation $O_1(\mathbf{v}), \ldots, O_n(\mathbf{v})$. For every type $i$, choose an additive clause $a_i^{\mathbf{v}}$ supporting $O_i(\mathbf{v})$. Define the benchmark item contribution

$$Q_{i,j}(\mathbf{v}) = \begin{cases} a^{\mathbf{v}}_{i,j}, & j \in O_i(\mathbf{v}), \\ 0, & j \notin O_i(\mathbf{v}). \end{cases}$$

Then

$$\sum_{i=1}^{n}\sum_{j\in M} Q_{i,j}(\mathbf{v}) = \mathrm{OPT}_1(\mathbf{v}). \tag{3}$$

Let the $N = nk$ arrivals be indexed by $t = 1, \ldots, N$, and write $i(t)$ for the type arriving at time $t$. Let $Q_{i,j}$ denote the random variable $Q_{i,j}(\mathbf{V})$ for an independent one-block benchmark sample $\mathbf{V} \sim F_1 \times \cdots \times F_n$.

For each item $j$, define deterministic continuation credits backward by

$$W_{N+1,j} = 0, \qquad W_{t,j} = W_{t+1,j} + \mathbb{E}\left[\left(Q_{i(t),j} - W_{t+1,j}\right)^+\right]. \tag{4}$$

Consider the dynamic item-price policy that posts price

$$p_{t,j} = W_{t+1,j}$$

for each item $j$ that is still available at time $t$.

**Proposition 5.1** (XOS continuation-price certificate)**.** *For every XOS instance, the policy defined by* (4) *satisfies*

$$\mathbb{E}[\mathrm{SW}_k] \geq \sum_{j \in M} W_{1,j}.$$

*Proof.* Let $R_t$ be the set of items remaining just before arrival $t$, let $S_t$ be the bundle purchased at time $t$, and let $u_t$ be the buyer's quasilinear utility. Let $\mathcal{H}_t$ be the actual history before time $t$, so $R_t$ and all posted prices are fixed after conditioning on $\mathcal{H}_t$. The current buyer's valuation is independent of $\mathcal{H}_t$.

First we prove the utility lower bound. Let $i = i(t)$ and let $\mathbf{V}$ be an independent benchmark profile. For this profile, let $O_i(\mathbf{V})$ be the benchmark bundle of type $i$, and let $a_i^{\mathbf{V}}$ be its supporting clause. Since the arriving buyer chooses a utility-maximizing bundle from $R_t$,

$$\begin{aligned}
\mathbb{E}[u_t \mid \mathcal{H}_t] &= \mathbb{E}_{v_i}\left[\max_{S \subseteq R_t}\left(v_i(S) - \sum_{j \in S} p_{t,j}\right)\right] \\
&= \mathbb{E}_{\mathbf{V}}\left[\max_{S \subseteq R_t}\left(V_i(S) - \sum_{j \in S} p_{t,j}\right)\right] \\
&\geq \mathbb{E}_{\mathbf{V}}\left[\sum_{j \in O_i(\mathbf{V}) \cap R_t} \left(a_{i,j}^{\mathbf{V}} - p_{t,j}\right)^+\right] \\
&= \sum_{j \in R_t} \mathbb{E}\left[\left(Q_{i(t),j} - p_{t,j}\right)^+\right].
\end{aligned}$$

The inequality uses the subset of the benchmark bundle consisting of exactly those items with positive supported surplus; the supporting-clause property implies that the buyer's value for that subset is at least the sum of the corresponding clause coefficients.

Now define the potential just before time $t$ by

$$\Phi_t = \sum_{j \in R_t} W_{t,j}.$$

Because $p_{t,j} = W_{t+1,j}$, the revenue from the purchased set plus the next potential is exactly

$$\sum_{j \in S_t} p_{t,j} + \Phi_{t+1} = \sum_{j \in S_t} W_{t+1,j} + \sum_{j \in R_t \setminus S_t} W_{t+1,j} = \sum_{j \in R_t} W_{t+1,j}.$$

Using the utility lower bound and the recursion (4),

$$\begin{aligned}
\mathbb{E}\left[v_{i(t)}(S_t) + \Phi_{t+1} \mid \mathcal{H}_t\right] &= \mathbb{E}\left[u_t + \sum_{j \in S_t} p_{t,j} + \Phi_{t+1} \mid \mathcal{H}_t\right] \\
&\geq \sum_{j \in R_t} \mathbb{E}\left[\left(Q_{i(t),j} - W_{t+1,j}\right)^+\right] + \sum_{j \in R_t} W_{t+1,j} \\
&= \sum_{j \in R_t} W_{t,j} = \Phi_t.
\end{aligned}$$

Telescoping over $t = 1, \ldots, N$ and using $\Phi_{N+1} = 0$ gives $\mathbb{E}[\mathrm{SW}_k] \geq \Phi_1 = \sum_j W_{1,j}$. □

**Remark 5.2** (Why Proposition 5.1 is not a full solution)**.** *For additive valuations, the correct item variables are the actual item values $X_{i,j}$, and a buyer purchases all items with positive item surplus. That is what permits the exact direct sum of single-item threshold processes in Theorem 4.1. In Proposition 5.1, the variables $Q_{i,j}$ are instead benchmark supporting-clause contributions. They are useful for analysis, but they are not item values that the mechanism can independently accept. A worst-case doubly-exponential lower bound on $\sum_j W_{1,j}$ relative to $\mathbb{E}[\mathrm{OPT}_1]$ is not known from this recursion, and this certificate should be viewed as a verified analysis tool rather than as a resolution of the XOS/submodular open problem.*

# 6 A limitation of the balanced-price coefficient template

The proof of the known $O(\log(1/\varepsilon))$ XOS/submodular bound uses balanced item prices based on expected optimal item contributions. The next lemma verifies a small algebraic fact: if one stays within the usual revenue-plus-surplus coefficient template, merely retuning the block multipliers cannot give double-exponential convergence. This is a limitation of that proof template, not a lower bound against all dynamic posted-price mechanisms.

Suppose an item has expected benchmark contribution $a_j$. In block $b$ a balanced-price proof posts $p_{b,j} = \alpha_b a_j$ with $0 \le \alpha_b \le 1$. Write $\delta_b = 1 - \alpha_b$ and $S_m = \sum_{b=1}^{m} \delta_b$. In the standard revenue-plus-surplus accounting, an item sold in block $m$ receives coefficient

$$C_m = \alpha_m + \sum_{b<m} (1 - \alpha_b) = 1 - \delta_m + S_{m-1},$$

and an item that survives all $k$ blocks receives coefficient

$$C_\infty = S_k.$$

The proof certifies the minimum of these coefficients.

**Lemma 6.1** (Optimal coefficient in the balanced-price template)**.** *For every $0 \le \alpha_1, \ldots, \alpha_k \le 1$,*

$$\min\{C_1, \ldots, C_k, C_\infty\} \le 1 - 2^{-k}.$$

*The bound is tight, attained by*

$$\alpha_b = 1 - 2^{-(k+1-b)} \qquad (b = 1, \ldots, k).$$

*Proof.* Let $\gamma = \min\{C_1, \ldots, C_k, C_\infty\}$. Since $C_m = 1 - \delta_m + S_{m-1} \ge \gamma$,

$$\delta_m \le 1 + S_{m-1} - \gamma.$$

Thus

$$S_m = S_{m-1} + \delta_m \le 1 + 2S_{m-1} - \gamma.$$

Starting from $S_0 = 0$, induction gives

$$S_k \le (2^k - 1)(1 - \gamma).$$

But $C_\infty = S_k \ge \gamma$, so

$$\gamma \le (2^k - 1)(1 - \gamma),$$

which rearranges to $\gamma \le 1 - 2^{-k}$.

For tightness, set $\delta_b = 2^{-(k+1-b)}$. Then

$$S_{m-1} = \sum_{b=1}^{m-1} 2^{-(k+1-b)} = 2^{-(k+1-m)} - 2^{-k}.$$

Therefore

$$C_m = 1 - 2^{-(k+1-m)} + 2^{-(k+1-m)} - 2^{-k} = 1 - 2^{-k}$$

for every $m$, and also

$$C_\infty = S_k = \sum_{b=1}^{k} 2^{-(k+1-b)} = 1 - 2^{-k}.$$

$\square$

# 7 Why the additive proof does not extend verbatim

The central identity in the additive proof is

$$\arg\max_{S \subseteq R} \left\{ \sum_{j \in S} X_{i,j} - \sum_{j \in S} p_j \right\} = \{j \in R : X_{i,j} > p_j\},$$

up to the zero-surplus tie convention. This identity is false for non-additive XOS and submodular valuations.

A two-item unit-demand valuation already shows the obstruction. Let $M = \{a, b\}$ and

$$v(\{a\}) = v(\{b\}) = v(\{a, b\}) = 1.$$

At prices $p_a = p_b = 1/3$, both singleton surpluses are positive, but buying both items gives utility $1 - 2/3 = 1/3$, whereas buying either singleton gives utility $1 - 1/3 = 2/3$. Thus a utility-maximizing unit-demand buyer buys at most one of the two items. Itemwise thresholds therefore do not induce independent itemwise stopping rules.

This is precisely the missing step in the full XOS/submodular problem. The continuation-price certificate in Proposition 5.1 recovers a valid itemwise accounting inequality, and Lemma 6.1 explains why the usual balanced-price coefficient accounting stops at $1 - 2^{-k}$. Neither result restores the separability of actual demand. A full solution would need a new mechanism or a new analysis that overcomes this nonseparable-demand obstruction.

# 8 Conclusion

The verified mathematical conclusion is

$$\boxed{\mathrm{CC}^{\mathrm{dyn}}_{\mathrm{additive}}(1 - \varepsilon) = \Theta(\log\log(1/\varepsilon)).}$$

This is a sharp theorem for a canonical subclass of XOS and submodular combinatorial auctions. It is partial progress toward Problem W4408830023, not a solution of the full problem. The additive theorem follows because additive demand decomposes into independent itemwise decisions. For general XOS or submodular valuations, demand coupling remains the core unresolved issue.

## References


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